

Intermediate + Hard Level Questions with Explanation

1. Print Numbers from 1 to 20

Question

Write a program to print numbers from 1 to 20 using a while loop.

Explanation

- Start from 1
 - Print the number
 - Increase the value by 1 each time
 - Stop when number becomes greater than 20
-

2. Print Even Numbers

Question

Write a program to print all even numbers between 1 and 50 using a while loop.

Explanation

- Even numbers are divisible by 2
 - Use loop from 1 to 50
 - Check condition using `if num % 2 == 0`
-

3. Sum of Natural Numbers

Question

Write a program to calculate the sum of first n natural numbers.

Explanation

Example:

$$1 + 2 + 3 + 4 + 5 = 15$$

- Take value of n
 - Use loop from 1 to n
 - Add each number into a sum variable
-

4. Reverse a Number

Question

Write a program to reverse a number using a while loop.

Explanation

Example:

1234 → 4321

Steps:

- Take last digit using % 10
 - Add digit into reverse number
 - Remove last digit using // 10
-

5. Count Digits in a Number

Question

Write a program to count total digits in a number.

Explanation

Example:

56789 → 5 digits

Steps:

- Use loop until number becomes 0
- Remove one digit each time

- Increase counter by 1
-

6. Palindrome Number

Question

Write a program to check whether a number is palindrome or not.

Explanation

Palindrome means same from both sides.

Example:

121 → Palindrome

1331 → Palindrome

Steps:

- Reverse the number
 - Compare with original number
-

7. Armstrong Number

Question

Write a program to check whether a number is Armstrong number.

Explanation

Example:

153

Calculation:

$$1^3 + 5^3 + 3^3 = 153$$

If sum equals original number → Armstrong number

8. Password Checker

Question

Write a program to continuously ask password until correct password is entered.

Explanation

- Store one correct password
 - Use while loop
 - Keep asking user input
 - Stop when password matches
-

9. Guessing Game

Question

Write a program for number guessing game.

Explanation

- Store one secret number
 - Ask user to guess
 - If wrong → ask again
 - If correct → print success message
-

10. Multiplication Table

Question

Write a program to print multiplication table of a number.

Explanation

Example for 5:

$$5 \times 1 = 5$$

$$5 \times 2 = 10$$

...

- Start from 1

- Multiply each value with given number
 - Repeat till 10
-

11. Fibonacci Series

Question

Write a program to print Fibonacci series.

Explanation

Series:

0 1 1 2 3 5 8 13 ...

Each next number is sum of previous two numbers.

12. Factorial of Number

Question

Write a program to find factorial of a number.

Explanation

Example:

$$5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$$

- Multiply numbers one by one using loop
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13. Menu Driven Calculator

Question

Create calculator using while loop and if elif else.

Explanation

Menu Example:

1. Add
2. Subtract

3. Multiply

4. Divide

5. Exit

- User selects option
 - Perform operation
 - Continue until exit option chosen
-

14. ATM System

Question

Create simple ATM program.

Explanation

Options:

- Check balance
- Deposit money
- Withdraw money
- Exit

Use loop so menu repeats again and again.

15. Sum Until User Enters 0

Question

Take numbers from user continuously and stop when user enters 0.

Explanation

- Add all numbers into sum
 - When input becomes 0 → stop loop
 - Print final sum
-

16. Leap Year Checker Repeatedly

Question

Check leap year repeatedly until user types "stop".

Explanation

- Ask year from user
- Check leap year condition
- Continue until "stop" entered

Leap year condition:

- Divisible by 4
 - Not divisible by 100 OR divisible by 400
-

17. Star Pattern

Question

Print star pattern using while loop.

Explanation

Pattern:

```
*  
**  
***  
****  
*****
```

- Start from 1 star
 - Increase stars each iteration
-

18. Prime Numbers Between 1 to 100

Question

Print all prime numbers from 1 to 100.

Explanation

Prime number:

- Divisible only by 1 and itself

Example:

2, 3, 5, 7, 11 ...

Use nested loop to check divisibility.

19. Student Marks Average

Question

Keep taking student marks until negative number entered.

Explanation

- Add all marks
- Count total students
- Stop on negative input
- Find average:

average = total / count

20. Login System with 3 Attempts

Question

Create login system with maximum 3 attempts.

Explanation

- Store correct username and password
- Ask user input
- If correct → Login successful

- If wrong → decrease attempts
- After 3 wrong attempts → block account